Lab 3 — Ticket Resolution-Time Regression

Completed solution using the supplied lab_ticket_resolution.csv dataset

1. Dataset inspection

The dataset contains **240 rows and 6 columns**. There are no missing values. ticket_id is an object/string identifier with 240 unique values; complexity_score and lines_changed are integers; team_experience_years and resolution_hours are floats; reopened is an integer binary column.

Statistic	complexity_score	lines_changed	team_experience_years	reopened	resolution_hours
count	240.000	240.000	240.000	240.000	240.000
mean	5.492	54.821	6.105	0.233	19.361
std	2.980	26.145	3.391	0.424	10.035
min	1.000	1.000	0.500	0.000	0.500
25%	3.000	33.000	3.300	0.000	11.015
50%	5.500	54.000	5.800	0.000	19.610
75%	8.000	74.000	9.200	0.000	27.123
max	10.000	113.000	12.000	1.000	42.500

First five rows:

ticket_id	complexity_score	lines_changed	team_experience_years	reopened	resolution_hours
T0001	6	67	5.6	1	24.77
T0002	9	72	11.5	0	33.43
T0003	6	49	3.3	0	17.82
T0004	5	50	9.6	1	19.54
T0005	3	33	6.4	1	11.04

2. Target, features, and prediction-time assumptions

Target y: resolution_hours. **Features X:** complexity_score and team_experience_years. ticket_id is excluded because it is only a row identifier and its arbitrary sequence should not predict resolution time. lines_changed is excluded because it is normally known during or after the fix; using it to predict resolution time before work begins would leak future information. reopened is excluded because reopening status is generally known only after an initial resolution. This assumes the prediction is made when the ticket first arrives. If the business prediction time differs, reconsider the feature list.

3. Split and evaluation workflow

An 80/20 train/test split was created with random_state=42. The test set was reserved for final metrics. Five-fold shuffled cross-validation (random_state=42) on the training set was used to compare polynomial degrees and tune Ridge/Lasso alpha. Metrics are MAE and RMSE in hours (lower is better) and R^2 (higher is better).

4–7. Test-set model results

Model	MAE (hours)	RMSE (hours)	R^2
Ridge (alpha=0.001)	2.906	3.572	0.884
Linear regression	2.906	3.572	0.884
Lasso (alpha=0.001)	2.906	3.572	0.884
Polynomial degree 2	3.020	3.741	0.873
Polynomial degree 3	3.140	3.949	0.858

Model	MAE (hours)	RMSE (hours)	R^2
Mean baseline	9.114	10.598	-0.021

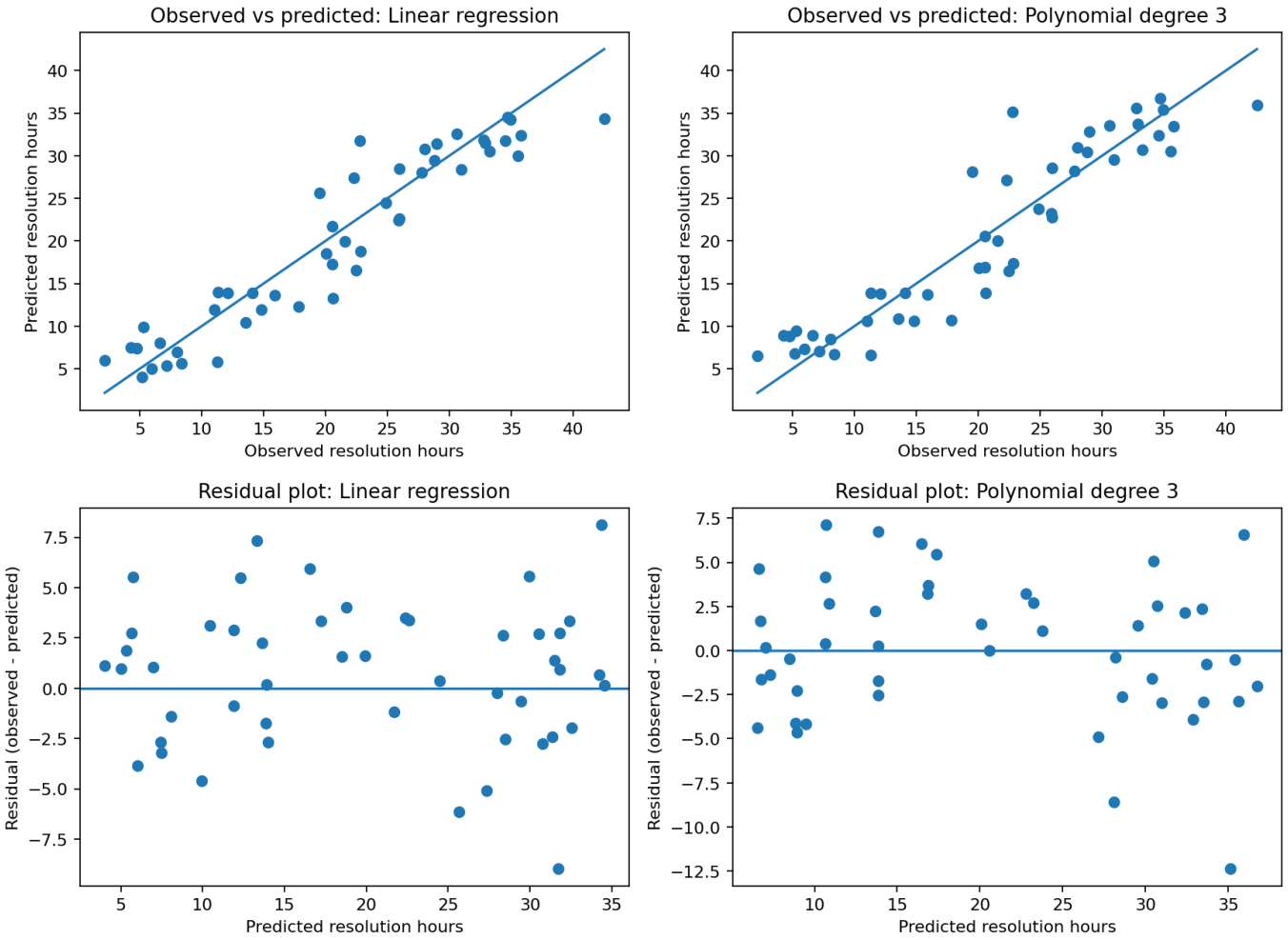
Ridge alpha selected by training-only CV: **0.001** (mean CV MAE 3.168 hours). Lasso alpha selected by training-only CV: **0.001** (mean CV MAE 3.168 hours). Both select the smallest tested alpha, so the grid suggests little regularization is needed for these two features.

Polynomial degree CV comparison

Degree	Mean 5-fold CV MAE (hours)	Std. dev.
2	3.131	0.310
3	3.162	0.188

Degree 2 performs slightly better than degree 3 in cross-validation, but both are worse than the linear model's test performance here. Increasing degree did not improve the evidence of generalization; degree 3 is a more flexible model and may fit noise more easily.

8. Diagnostic plots



In observed-versus-predicted plots, points close to the diagonal indicate better predictions. In residual plots, a random band around zero is desirable. Look for systematic curves, widening/narrowing spread, and points far from the rest. With a small synthetic test set, these plots are diagnostic clues, not proof.

9–17. Recommendation and analysis questions

9. Recommendation

Choose ordinary linear regression as the practical default. It ties with Ridge/Lasso to the displayed precision, has test MAE about 2.906 hours, RMSE about 3.572 hours, and R^2 about 0.884. It is simpler than the polynomial models, and the tested polynomial degrees did not improve validation MAE. Ridge/Lasso are essentially equivalent at the selected very small alpha. Before deployment, collect real historical data, verify feature availability, test over time and across teams, and monitor prediction errors.

10. Why is the mean baseline useful?

It provides a simple reference: it predicts the average resolution time for every ticket without using features. A useful model should beat this naive benchmark. Here the baseline test MAE is about 9.114 hours and R^2 is about -0.021, much worse than the feature-based models.

11. Lowest MAE and RMSE; why can they differ?

Ridge (alpha=0.001) has the lowest values in the displayed test table, but only by a negligible amount; linear regression is effectively tied. MAE averages absolute errors, while RMSE squares errors before averaging and therefore penalizes large errors more heavily. A model that reduces a few large misses can improve RMSE without having the lowest MAE.

12. Did polynomial features improve validation? Overfitting evidence?

No. Degree 2 has mean training-fold CV MAE 3.131 hours and degree 3 has 3.162; degree 2 is slightly better. Their test MAEs (about 3.020 and 3.140) are worse than the linear model. The more complex degree-3 model performing worse is consistent with unnecessary complexity/possible overfitting, though a definitive diagnosis requires more data and repeated validation.

13. How did regularization affect performance and coefficients?

At alpha=0.001, Ridge and Lasso perform almost identically to ordinary linear regression. Small alpha shrinks coefficients only slightly; larger alpha values worsen CV MAE in this grid, especially for Lasso, which can shrink coefficients toward or to zero. Regularization trades some flexibility/fit for reduced coefficient magnitude and potentially better generalization.

14. Why scale Ridge/Lasso features?

Ridge penalizes the sum of squared coefficients and Lasso the sum of absolute coefficients. If features use different numeric scales, the penalty can affect coefficients unevenly. StandardScaler puts features on comparable scales so the penalty is applied more fairly. Scaling is fitted within the pipeline to avoid leakage from validation/test data.

15. Potential leakage and handling

lines_changed is a likely leakage risk because it describes code changes made during the fix, information unavailable when predicting the time required before work starts. reopened may also be post-outcome information. Both are excluded under the stated prediction-time assumption; ticket_id is excluded as a non-predictive identifier.

16. Is random splitting appropriate for multi-year or repeated team/customer data?

Not necessarily. A random split can put records from the same team/customer or later time periods in both train and test sets, making performance look overly optimistic. For future prediction, use a time-based split (train on earlier tickets, test on later tickets). For repeated customers/teams, use group-aware splitting so groups do not cross the train/test boundary, or report both evaluations.

17. Additional tests before deployment

Evaluate on a later time period and on unseen teams/customers; compare against operational baselines; inspect residuals and large-error cases; check errors by complexity level/team and for bias; validate feature definitions and data quality; test missing/unusual inputs and

distribution shift; assess prediction intervals/uncertainty; confirm privacy/security and workflow impact; then monitor drift and retrain only under a documented process.

Reproducibility note

The data is synthetic, as stated in the lab brief. These results demonstrate a workflow and should not be treated as evidence of real-world organizational performance. The included Python script reruns inspection, model selection, metrics, and diagnostic plotting when the CSV is in the same folder.